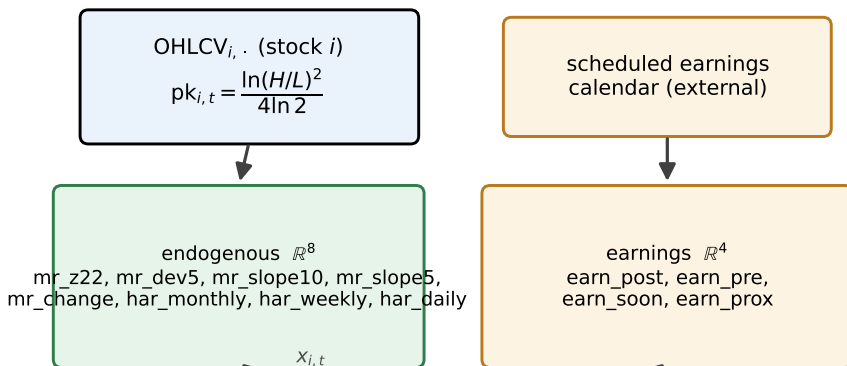


$XGB: x \in \mathbb{R}^8$ (own) · $XGB+E: \mathbb{R}^{12}$ (own+E) · $XGB+E+LG: +$ leaf-graph smoothing



Gamma-loss gradient boosting (XGBoost)

$$F(x) = \sum_{m=1}^{300} \eta f_m(x), \quad f_m: \text{regression tree}$$

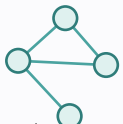
(31 leaves, $\eta = 0.05$) · reg:gamma loss = QLIKE+const
 3-seed ensemble · trained per market: S&P 500 · HOSE

per-stock forecast $\hat{\sigma}_{i,t+h}^2$
 $h \in \{1, 5, 10, 22\}$

$\hat{y}_i$ (all stocks)

leaf-cooccurrence graph smoothing

(post-prediction step)


 same-day cross-section;
 kNN on shared
 XGBoost leaves ($k = 10$)

$$\hat{y}_i^{sm} = (1 - \alpha) \hat{y}_i + \alpha \bar{\hat{y}}_{\text{kNN}(i)}$$

leaf-graph smoothing ($\alpha \in [0, 1]$ fit on validation)

prediction smoothing — not a GNN, not a feature

→ $\hat{y}_{i,t+h}^{sm}$: final smoothed forecast